

Diffraction Applications, Thin-Film Interference, and Anti-Reflection Coatings

PHYS 2203 — Introductory Physics III

September 18, 2026

Overview

After a brief recap of single-slit diffraction, we combine it with two-slit interference — the fringes riding under a diffraction envelope — and then finish the diffraction topic begun last lecture: circular-aperture diffraction and the Rayleigh criterion for the resolving power of telescopes and microscopes. We then turn to **thin-film interference** — light reflected from the two surfaces of a thin film, whose path difference and reflection phase shifts explain the colors of soap bubbles and oil films and the design of anti-reflection coatings. An optional appendix derives the circular-aperture (Airy) result using Bessel functions.

1 Recap: Single-Slit Diffraction

Last lecture we modeled a slit of width a as the $N \rightarrow \infty$ limit of N point sources (Huygens's principle) and found the intensity

$$\boxed{I \sim \frac{\sin^2 \alpha}{\alpha^2}}, \quad \alpha = \frac{\pi a}{\lambda} \sin \theta, \quad (1)$$

a bright central maximum flanked by progressively weaker side peaks. The dark fringes fall at $\alpha = m\pi$, that is

$$a \sin \theta = m\lambda, \quad m = \pm 1, \pm 2, \dots, \quad (2)$$

and the central peak (between the $m = \pm 1$ minima) has full angular width

$$\Delta\theta = 2 \sin^{-1} \left(\frac{\lambda}{a} \right) \approx \frac{2\lambda}{a}. \quad (3)$$

The narrower the slit, the wider the spread, so diffraction is most pronounced when $a \sim \lambda$ (note that there are no solutions for $a < \lambda$, so there are no minima and the slit behaves as a point source). This lecture extends the same idea to a *circular* aperture and works out its consequences for the resolution of optical instruments.

2 Interference Combined with Diffraction

The two effects now combine. Real slits have a finite width, so a two-slit pattern carries *both*: the sharp interference fringes set by the slit *separation* d , and the broad single-slit diffraction envelope

set by the slit *width* a . Because both slits have the same width, they share one common diffraction envelope, and the full pattern is simply the product of the two results:

$$I(\theta) \propto \underbrace{\left(\frac{\sin \alpha}{\alpha}\right)^2}_{\text{diffraction (width } a)} \times \underbrace{\cos^2 \frac{\delta}{2}}_{\text{interference (separation } d)}, \quad \alpha = \frac{\pi a}{\lambda} \sin \theta, \quad \delta = \frac{2\pi}{\lambda} d \sin \theta. \quad (4)$$

The sharp fringes ride underneath the broad envelope (Figure 1). One consequence: if d/a is an integer, an interference maximum lands exactly on a diffraction zero and disappears — a *missing order*.

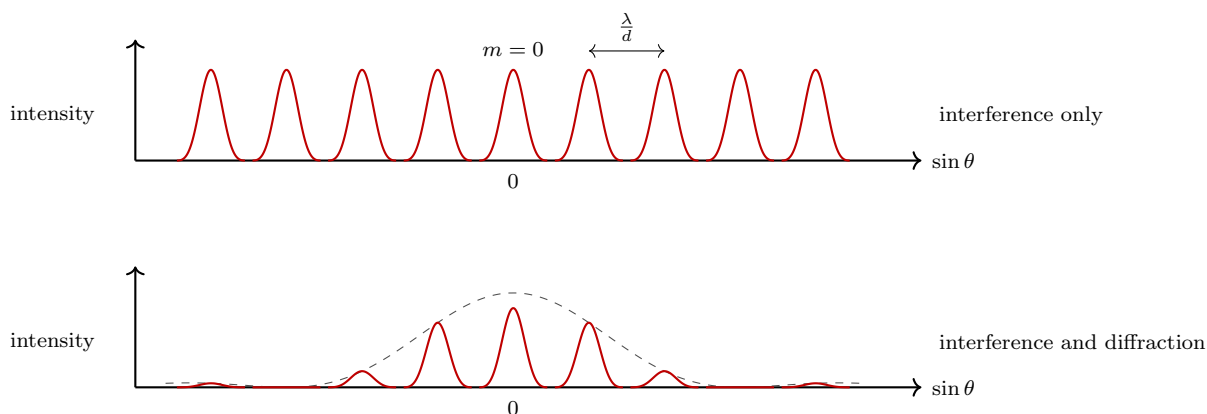


Figure 1: *Top*: idealized two-slit interference (equal-height fringes, spacing λ/d). *Bottom*: the physical pattern — the same fringes modulated by the single-slit diffraction envelope (dashed). Here $d/a = 3$, so every third order ($m = \pm 3, \pm 6, \dots$) falls on a diffraction zero and is missing.

3 Circular Apertures

For a **circular aperture**, the first intensity minima occur when

$$\boxed{a \sin \theta = 1.220 \lambda,} \quad \text{where } a = \text{diameter of aperture} \quad (5)$$

to be contrasted with $a \sin \theta = \lambda$ for a slit. The extra factor of 1.220 comes from the circular geometry.

The derivation requires knowledge of Bessel functions and is too advanced for this course. It is worked out in an appendix at the end of this lecture, for the curious.

4 Resolution of Optical Instruments

Diffraction often limits the resolution of a telescope or microscope. Light from a distant object passing through the circular aperture of the objective lens cannot form a perfect point image; it forms a diffraction blur of finite angular size.

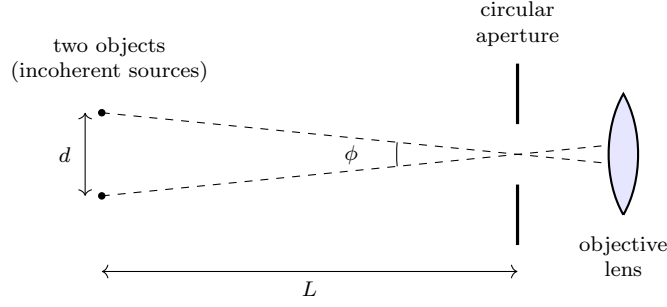


Figure 2: Two object points separated by d at distance L subtend an angle ϕ at the circular aperture in front of the objective lens. The aperture diffracts each point into a blur of finite angular size.

The angle subtended by the object satisfies

$$\tan \frac{\phi}{2} = \frac{d}{2} \frac{1}{L} \quad \Rightarrow \quad d \approx \phi L \quad (\text{small angles}). \quad (6)$$

The Rayleigh criterion

Consider **two objects** acting as incoherent sources. Each forms, on the image plane of the lens, a diffraction pattern whose central peak has angular half-width θ_1 set by the first minimum,

$$a \sin \theta_1 = 1.220 \lambda \quad (\text{first minimum}), \quad 2\theta_1 = 2 \sin^{-1} \left(\frac{1.220 \lambda}{a} \right). \quad (7)$$

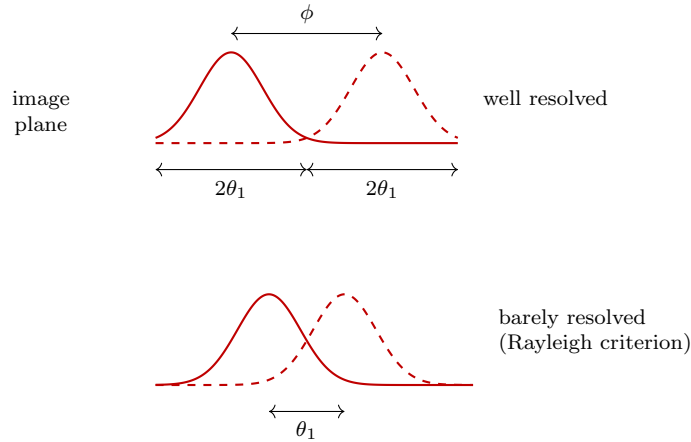


Figure 3: Two incoherent point sources imaged through a circular aperture. Top: when their angular separation ϕ comfortably exceeds the peak half-width, the images are well resolved. Bottom: at the Rayleigh criterion the peak of one image falls on the first minimum of the other ($\phi = \theta_1$); they are barely resolved.

The objects are resolvable when their angular separation is at least the diffraction half-width:

$$\phi \geq \theta_1 \quad \Rightarrow \quad \boxed{\phi \geq \sin^{-1} \left(\frac{1.220 \lambda}{a} \right)}. \quad (8)$$

Maximizing resolution

How can we maximize the resolution of a microscope or telescope? Three strategies follow directly from the criterion:

1. **Decrease L** — place the aperture closer to the object.
2. **Increase a** — use a larger aperture and objective lens.
3. **Decrease λ** — use blue light instead of red, or use accelerated electrons instead of light.

5 Thin-Film Interference

Consider **normal incidence only** on a thin film. Light incident from material 1 partially reflects at the first boundary and partially transmits into material 2 (the film, thickness d). The transmitted wave then partially reflects at the second boundary and emerges back into material 1. The two reflected waves — one from the first boundary, one from the second — recombine and interfere.

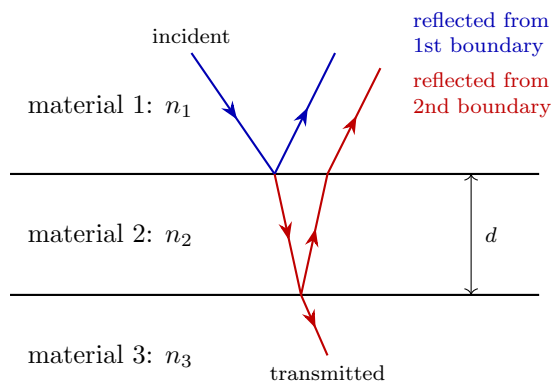


Figure 4: Normal incidence on a thin film of thickness d . The wave reflected from the first boundary interferes with the wave reflected from the second boundary. (The rays are drawn at an angle for clarity.)

Important factors

1) Path difference. The wave reflected from the second boundary travels an extra distance $2d$ (down and back through the film) relative to the wave reflected from the first boundary:

$$\text{path difference between reflected waves} = 2d, \quad \text{phase difference} = 2d \frac{2\pi}{\lambda_2}, \quad (9)$$

where λ_2 is the wavelength *inside* the film (material 2).

2) Wavelength and speed depend on the medium. The speed of light in a medium of refractive index n is

$$v = \frac{c}{n}, \quad (10)$$

where c is the speed of light in vacuum and n is the index of refraction. Since the frequency f is independent of the medium,

$$v = f\lambda_n \implies \lambda_n = \frac{v}{f} = \frac{c}{f} \frac{1}{n} \implies \boxed{\lambda_n = \frac{\lambda}{n}} \quad (11)$$

where λ is the wavelength of light in vacuum.

3) Phase change on reflection. If $n_2 > n_1$, then the reflected wave undergoes a phase change of π (reflection off a higher-index, “denser” medium). If $n_2 < n_1$, there is no phase change.

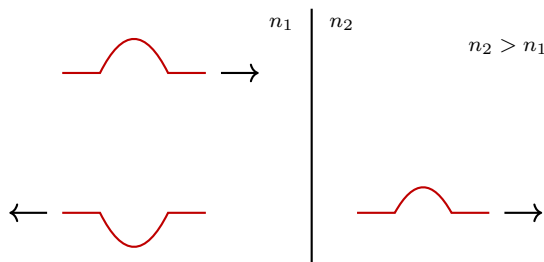


Figure 5: A wave reflecting off a higher-index medium ($n_2 > n_1$) is inverted: it acquires a phase change of π . The transmitted wave is not inverted.

Where the phase change comes from

Recall from Lecture 4. Matching the tangential $\vec{E}$ and $\vec{B}$ across the boundary at normal incidence gave the **reflection** and **transmission** coefficients

$$\boxed{r \equiv \frac{E_r}{E_i} = \frac{n_1 - n_2}{n_1 + n_2}, \quad t \equiv \frac{E_t}{E_i} = \frac{2n_1}{n_1 + n_2}.} \quad (12)$$

The phase change on reflection is simply the *sign* of r . If $n_2 > n_1$ (reflection off the denser medium), then $r < 0$: the reflected field comes out with its sign flipped, and for a sinusoid a sign flip *is* a phase shift of π ($-\sin \theta = \sin(\theta + \pi)$). If $n_2 < n_1$, then $r > 0$ and there is no flip. The transmitted coefficient $t > 0$ always, so the transmitted wave is never inverted.

6 Example: Anti-Reflection Coating

A thin film with $n_1 < n_2 < n_3$ is deposited on glass to **minimize reflection**. Light of vacuum wavelength $\lambda_1 = 600$ nm is incident from air ($n_1 = 1$) onto a film ($n_2 = 1.35$) on glass ($n_3 = 1.5$).

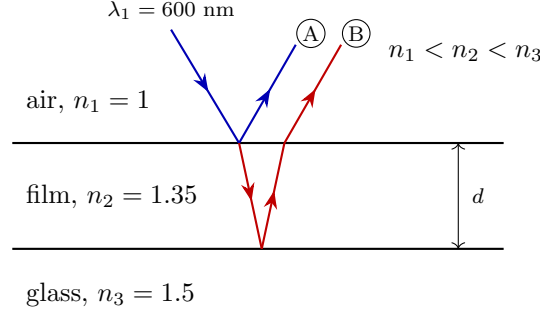


Figure 6: Anti-reflection coating. Reflections A (air/film) and B (film/glass) interfere. The film thickness d is chosen to make them cancel. (The rays are drawn at an angle for clarity.)

Question. How thick should the film be in order to minimize reflection?

Since $n_1 < n_2$, ray A (reflection at the first boundary) gets a phase shift of π . Since $n_2 < n_3$, ray B (reflection at the second boundary) also gets a phase shift of π . Collecting the path difference and both reflection shifts,

$$\delta = \underbrace{2d \frac{2\pi}{\lambda_2}}_{\text{path difference}} + \underbrace{\pi}_{\text{1st reflection}} + \underbrace{\pi}_{\text{2nd reflection}} = 2d \frac{2\pi}{\lambda_2} + 2\pi. \quad (13)$$

Both reflections add a phase of π , so together they contribute 2π . A phase shift of 2π is a full cycle — it returns each wave to exactly where it started — so it has no effect on the interference and drops out, leaving the effective phase difference

$$\delta = 4\pi \frac{d}{\lambda_2}. \quad (14)$$

The reflected rays **destructively interfere** when $\delta = (\text{odd integer})\pi$:

$$4\pi \frac{d}{\lambda_2} = m\pi, \quad m \text{ odd}, \quad (15)$$

$$d = \left(\frac{m}{4}\right) \lambda_2 = \frac{\text{odd integer}}{4} \lambda_2. \quad (16)$$

Here $\lambda_2 = \lambda_1/n_2$ is the wavelength inside the film, so

$$d = \frac{\text{odd integer}}{4} \frac{600 \text{ nm}}{1.35} = (\text{odd integer})(111 \text{ nm}). \quad (17)$$

The thinnest such coating ($m = 1$) is about 111 nm.

7 Example: Air Gap Between Two Glass Plates

Now consider an air film ($n_2 = 1.0$) sandwiched between two glass plates ($n_1 = n_3 = 1.5$), illuminated by light of vacuum wavelength $\lambda_{\text{air}} = 600 \text{ nm}$.

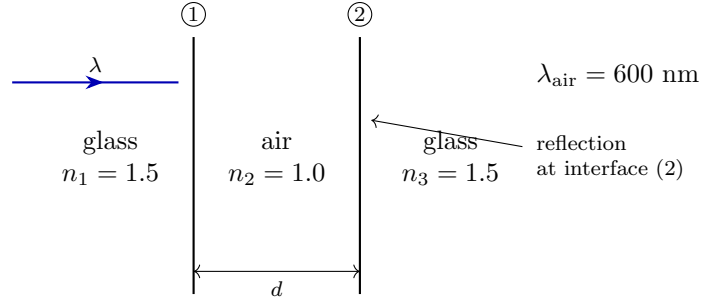


Figure 7: An air gap of thickness d between two glass plates. Reflection at interface (1) is glass→air (no phase change); reflection at interface (2) is air→glass (phase change π).

At interface (1) the light reflects glass→air ($n_2 < n_1$): *no* phase change. At interface (2) it reflects air→glass ($n_3 > n_2$): a phase change of π . For minimum reflection,

$$\underbrace{2d \frac{2\pi}{\lambda_{\text{air}}}}_{\text{due to path difference}} + \underbrace{\pi}_{\text{reflection at interface (2)}} = (\text{odd integer}) \pi. \quad (18)$$

Writing this out,

$$4\pi \frac{d}{\lambda_{\text{air}}} + \pi = (\text{odd}) \pi, \quad (19)$$

$$d = \left[\frac{(\text{odd}) - 1}{4} \right] \lambda_{\text{air}} = \frac{\text{even}}{4} \lambda_{\text{air}}. \quad (20)$$

The minimum nonzero thickness for minimized reflection (even integer = 2) is

$$\boxed{d = \frac{2}{4} (600 \text{ nm}) = 300 \text{ nm}.} \quad (21)$$

Newton's rings. A curved glass surface resting on a flat plate forms an air gap of slowly varying thickness, producing concentric bright and dark interference fringes known as **Newton's rings**.

Other examples. The same thin-film interference produces the colors of *oil on water* and *soap bubbles*.

Optional: Deriving the Circular-Aperture Result (the Airy Pattern)

This appendix derives $a \sin \theta = 1.220 \lambda$. Nothing here is examinable; it is included for the curious. The only new inputs are four facts about the Bessel functions J_0 and J_1 , quoted without proof:

$$\begin{aligned} \text{(i)} \quad J_0(u) &= \frac{1}{2\pi} \int_0^{2\pi} e^{-iu \cos \varphi} d\varphi, & \text{(ii)} \quad \frac{d}{du} [u J_1(u)] &= u J_0(u), \\ \text{(iii)} \quad J_1(u) &\approx \frac{u}{2} \text{ for small } u, & \text{(iv)} \quad \text{the first nonzero root of } J_1 &\text{ is } u_1 = 3.8317. \end{aligned}$$

Setup. In the far field, the aperture element at transverse position x contributes an amplitude with phase factor $e^{-ikx \sin \theta}$, where θ is the observation angle — the continuum version of the phase differences used in the N -slit analysis of last lecture.

Warm-up: the slit, by direct integration. For a slit of width a the superposition is a single integral, and it reproduces the single-slit result of last lecture without any limit-taking:

$$E(\theta) \propto \int_{-a/2}^{a/2} e^{-ikx \sin \theta} dx = \frac{2 \sin\left(\frac{ka}{2} \sin \theta\right)}{k \sin \theta} = a \frac{\sin \alpha}{\alpha}, \quad \alpha = \frac{\pi a}{\lambda} \sin \theta,$$

so $I \propto \sin^2 \alpha / \alpha^2$, as before.

The circular aperture. For a circular aperture of radius $R = a/2$ the superposition is a double integral. In polar coordinates ($x = \rho \cos \varphi$, $dA = \rho d\rho d\varphi$), the φ integral is exactly fact (i) with $u = k\rho \sin \theta$:

$$E(\theta) \propto \int_0^R \rho d\rho \int_0^{2\pi} e^{-ik\rho \sin \theta \cos \varphi} d\varphi = 2\pi \int_0^R J_0(k\rho \sin \theta) \rho d\rho.$$

Substituting $u = k\rho \sin \theta$, so that $\rho d\rho = u du / (k \sin \theta)^2$, and writing $x \equiv kR \sin \theta$:

$$E(\theta) \propto \frac{2\pi}{(k \sin \theta)^2} \int_0^x u J_0(u) du = \frac{2\pi}{(k \sin \theta)^2} x J_1(x) = \pi R^2 \frac{2 J_1(x)}{x},$$

using fact (ii) in integrated form, $\int_0^x u J_0(u) du = x J_1(x)$, and $(k \sin \theta)^2 = x^2 / R^2$. The intensity is the **Airy pattern**:

$$I(\theta) = I_0 \left[\frac{2 J_1(x)}{x} \right]^2, \quad x = \frac{\pi a}{\lambda} \sin \theta.$$

The center and the first minimum. By fact (iii), $2J_1(x)/x \rightarrow 1$ as $x \rightarrow 0$: the center carries the peak intensity I_0 , just as $\sin \alpha / \alpha \rightarrow 1$ for the slit. The first zero of the intensity is the first nonzero root of J_1 , fact (iv):

$$\frac{\pi a}{\lambda} \sin \theta_1 = 3.8317 \quad \implies \quad a \sin \theta_1 = \frac{3.8317}{\pi} \lambda = 1.220 \lambda. \quad \checkmark$$

The mysterious factor 1.220 is nothing but the first root of J_1 in units of π — to be compared with the first root of $\sin$, which is exactly π , giving $a \sin \theta_1 = \lambda$ for the slit.